



Overview of Databases and Transaction Processing

Database Systems

- DB Systems :- Data, H/W, S/W, & Users
- Data : integrated, shared (multi-user) ,
- H/W : CPU, memory, secondary storage, N/W (client/server, host/terminal)
- S/W : DB server, DBMS
- Users : application programmers, end users, DB administrator (DBA)

What is a Database?

- Collection of data central to some enterprise
- Essential to operation of enterprise
 - Contains the only record of enterprise activity
- An asset in its own right
 - Historical data can guide enterprise strategy and interest to other enterprises
- State of database mirrors state of enterprise
 - Database is persistent

What is a Database Management System?

- A Database Management System (DBMS) is a program that manages a database:
 - Supports a high-level access language (e.g. SQL).
 - Application describes database accesses using that language.
 - DBMS interprets statements of language to perform requested database access.

What is a Data Model?

- Model is representation of a real world situation about which data is to be collected and stored in a database.
- A data model depicts the dataflow and logical interrelationships among different data elements.
- Data model consist of :
 - Objects : to model the structure of data.
 - Operators : to model it behavior.

Why Database?

- Compactness
- Speed
- Less drudgery
- Currency

Benefit of Database

- The data can be shared
- Redundancy can be reduced
- Inconsistency can be avoided
- Transaction support can be provided
- Integrity can be maintained (integrity constraint, business rules)
- Security can be enforced
- Standards can be enforced

Data Independent

- Different applications will require different views of the same data
- The DBA must have the freedom to change the physical representation or access technique in response to changing requirement, without having to modify existing applications

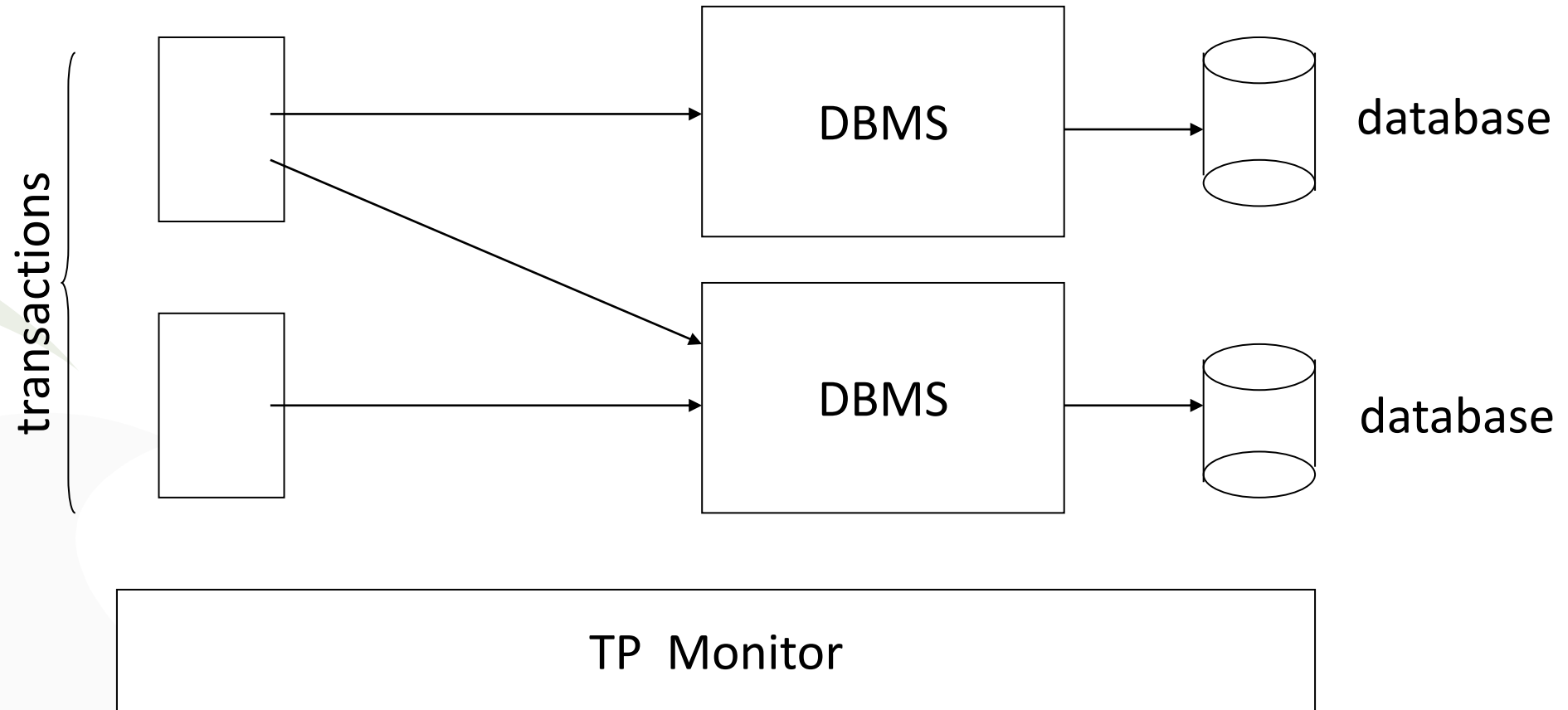
What is a Transaction?

- When an event in the real world changes the state of the enterprise, a transaction is executed to cause the corresponding change in the database state
 - With an on-line database, the event causes the transaction to be executed in real time
- In the context of a database, a transaction can be defined as a group of tasks (database operations). It is a unit of a program execution that accesses and possibly modifies various data objects (database tuples, relations).
- A transaction is a logical unit of database processing that includes one or more access operations:
 - Read - retrieval of information from database
 - Write - insert or update in the database, delete data from the database

What is a Transaction Processing System?

- Transaction execution is controlled by a TP monitor
 - Creates the abstraction of a transaction, analogous to the way an operating system creates the abstraction of a process
 - TP monitor and DBMS together guarantee the special properties of transactions
- A Transaction Processing System consists of TP monitor, databases, and transactions

Transaction Processing System



DB Systems: Then and Now

- Most generally is relational model using SQL - high-level view of data (may be NoSQL DB in some enterprise)
 - Older systems presented low-level view
- Might contain multimedia data
 - Older systems restricted to alphanumeric data
- On-line: database accessed at time of event
 - Older systems were off-line, batch

Systems: Then and Now (con't)

- Concurrent - multiple transactions execute simultaneously
 - Older systems processed transactions sequentially
- Distributed computation - different parts of the application execute on different computers
 - Older systems were centralized

Systems: Then and Now (con't)

- Distributed data - different parts of the data are stored in different databases on different computers
 - Older systems were centralized
- Heterogeneous - involves HW and SW modules from different manufacturers
 - Older systems were homogeneous
- Accessed by everyone (e.g., e-commerce)
 - Older systems restricted to trained personnel

Database System Requirements

- **High Availability:** on-line => must be operational while enterprise is functioning
- **High Reliability:** correctly tracks state, does not lose data, controlled concurrency
- **High Throughput:** many users => many transactions/sec
- **Low Response Time:** on-line => reduce users waiting time

System Requirements (con't)

- **Long Lifetime:** complex systems are not easily replaced
 - Must be designed so they can be easily extended as the needs of the enterprise change
- **Trusted Security:** sensitive information must be carefully protected since system is accessible to many users
 - Authentication, authorization, encryption

Roles in Design, Implementation, and Maintenance of a TPS

- **System Analyst** - specifies system using input from customer, provides complete description of functionality from customer's and user's point of view
- **Database Designer** - specifies structure of data that will be stored in database
- **Application Programmer** - implements application programs (transactions) that access data and support enterprise rules

Roles in Design, Implementation and Maintenance of a TPS (con't)

- **Database Administrator** - maintains database once system is operational: space allocation, performance optimization, database security
- **System Administrator** - maintains transaction processing system: monitors interconnection of HW and SW modules, deals with failures and congestion